

NTU SPACE DATA CENTER

NTU SPACE DYNAMICS

Software Module Test Report

Orbit • Attitude • Solar Environment • Power and Thermal • Access Analysis

Institute	College of Computing and Data Science, Nanyang Technological University
Author	Qin Jiahao
Reviewer	Liu Xiyuan
Issue date	17 August 2026

College of Computing and Data Science, Nanyang Technological University

Version Control

Abstract	This report defines the verification scope, procedures, expected results, acceptance criteria, and execution records for the orbit-dynamics module of the Space Data Center digital twin.			
Keywords	digital twin; orbit dynamics; attitude dynamics; Sun environment; access analysis; verification			
Change Record				
Revision	Change Request	Date	Author	Description
1.0	Initial issue	2026-08-17	Qin Jiahao	Orbit dynamics module test report based on the retained outline.

TABLE OF CONTENTS

1 Overview	1
1.1 Identification.....	1
1.2 Software and Document Overview.....	1
1.3 Abbreviations and Conventions.....	1
2 Test Content.....	2
3 Detailed Test Projects	4
3.1 Orbit Propagation and Coordinate Conversion	4
3.1.1 Test Project 1: TWOBODY	4
3.1.2 Test Project 2: J2.....	5
3.1.3 Test Project 3: SGP4.....	6
3.1.4 Test Project 4: HPOP	7
3.1.5 Test Project 5: OE2RV.....	8
3.1.6 Test Project 6: RV2OE.....	9
3.1.7 Test Project 7: ECI2ECEF	10
3.2 Attitude Kinematics and Dynamics	12
3.2.1 Test Project 8: YPR2EULER.....	12
3.2.2 Test Project 9: EULERDYNAMICS	13
3.3 Sun, Radiation, Thermal, and Power.....	15
3.3.1 Test Project 10: SUN POSITION ANALYTICAL	15
3.3.2 Test Project 11: SUN POSITION EPHEMERIS	16
3.3.3 Test Project 12: SUN BETA ANGLE	17
3.3.4 Test Project 13: SUN INTENSITY	18
3.3.5 Test Project 14: POWER CONSIDERING THERMAL	19
3.3.6 Test Project 15: SUN RADIATION PRESSURE.....	20
3.4 Access Analysis	21
3.4.1 Test Project 16: ACCESS PROPAGATION.....	21
3.4.2 Test Project 17: ACCESS INTERPOLATION	22
4 Test Content Sufficiency Analysis.....	24

5 Test Conditions and Requirements	24
6 Test Data.....	24
7 Test Summary	26

1 Overview

1.1 Identification

Platform name: SDCTwin.

Module under test: Orbit Dynamics Module.

Document title: Test Report for Orbit Dynamics Module of SDCTwin.

Document purpose: Define repeatable functional and numerical verification for the requested orbit, attitude, Sun, power, radiation-pressure, and access-analysis capabilities.

1.2 Software and Document Overview

This report is intended for system testers, orbital-dynamics developers, simulation integrators, and reviewers. It defines the test intent and execution record for 17 module functions. Because no runtime logs, build identifier, configuration manifest, or reference comparison output was supplied, the report does not assert that any case has passed.

1.3 Abbreviations and Conventions

Term	Definition
ECI	Earth-Centered Inertial; exact realization shall be identified in each run.
ECEF	Earth-Centered, Earth-Fixed; exact realization shall be identified in each run.
HPOP	High-Precision Orbit Propagator.
OE / RV	Orbital elements / Cartesian position and velocity.
RAAN	Right ascension of the ascending node.
YPR	Yaw-pitch-roll attitude convention, including a documented axis sequence.

2 Test Content

The verification scope contains 17 functions organized into four capability groups. Each function is traced to one detailed test case and one final summary record.

- Orbit propagation and coordinate conversion: TWOBODY, J2, SGP4, HPOP, OE2RV, RV2OE, and ECI2ECEF.
- Attitude kinematics and dynamics: YPR2EULER and EULERDYNAMICS.
- Sun, radiation, thermal, and power: SUN POSITION ANALYTICAL, SUN POSITION EPHEMERIS, SUN BETA ANGLE, SUN INTENSITY, POWER CONSIDERING THERMAL, and SUN RADIATION PRESSURE.
- Access analysis: ACCESS PROPAGATION and ACCESS INTERPOLATION.

Table 1. Functional Requirement and Test-Case Traceability

No.	Case ID	Function	Test Name	Planned Status
1	OD-001	TWOBODY	Two-Body Orbit Propagation	Not Executed
2	OD-002	J2	J2-Perturbed Orbit Propagation	Not Executed
3	OD-003	SGP4	SGP4 TLE Propagation	Not Executed
4	OD-004	HPOP	High-Precision Orbit Propagation	Not Executed
5	CV-001	OE2RV	Orbital Elements to Position and Velocity	Not Executed
6	CV-002	RV2OE	Position and Velocity to Orbital Elements	Not Executed
7	CV-003	ECI2ECEF	ECI to ECEF Coordinate Transformation	Not Executed
8	AD-001	YPR2EULER	Yaw-Pitch-Roll to Euler-Angle Conversion	Not Executed
9	AD-002	EULERDYNAMICS	Rigid-Body Euler Dynamics	Not Executed
10	SE-001	SUN POSITION ANALYTICAL	Analytical Sun Position	Not Executed
11	SE-002	SUN POSITION EPHEMERIS	Ephemeris-Based Sun Position	Not Executed
12	SE-003	SUN BETA ANGLE	Sun Beta-Angle Calculation	Not Executed
13	SE-004	SUN INTENSITY	Solar Intensity and Eclipse Factor	Not Executed

No.	Case ID	Function	Test Name	Planned Status
14	SE-005	POWER CONSIDERING THERMAL	Solar Power with Thermal Effects	Not Executed
15	SE-006	SUN RADIATION PRESSURE	Solar Radiation Pressure Acceleration	Not Executed
16	AC-001	ACCESS PROPAGATION	Access Interval Propagation	Not Executed
17	AC-002	ACCESS INTERPOLATION	Access State and Event Interpolation	Not Executed

3 Detailed Test Projects

The following records use the same execution structure as the reference outline. The numerical thresholds are default verification gates and shall be tightened when an approved module requirement defines a stricter value.

3.1 Orbit Propagation and Coordinate Conversion

3.1.1 Test Project 1: TWOBODY

This project verifies the Two-Body Orbit Propagation capability of the Space Data Center digital twin orbit-dynamics module.

Table 2. OD-001 - Two-Body Orbit Propagation Test Case

Case ID	OD-001	Case Name	Two-Body Orbit Propagation
Test Objective	Verify Keplerian two-body propagation, state continuity, reversibility, and conservation of the two-body invariants.		
Preconditions	• The digital twin orbit-dynamics module is prepared with only central-body point-mass gravity enabled. • The gravitational parameter and unit system are recorded in the run manifest. • An independent universal-variable or Kepler-equation solver is available as the reference oracle.		
Input Data	• Nominal LEO at 2025-01-01T00:00:00 UTC: $a = 7000$ km, $e = 0.001$, $i = 98$ deg, RAAN = 40 deg, argument of periapsis = 30 deg, true anomaly = 10 deg. • Additional circular, eccentric, equatorial, and retrograde states; propagation spans of one period and 24 h; output step of 60 s.		
Procedure	1. Load each initial state and confirm that only the two-body force is active. 2. Propagate forward and compare every sampled position and velocity with the independent two-body reference. 3. Compute specific mechanical energy and angular-momentum magnitude at every sample. 4. Propagate the final state backward to the initial epoch and compare with the original state.		
Expected Results	• The generated trajectory is continuous and has the expected Keplerian period and geometry. • Energy and angular momentum remain constant within numerical integration tolerance. • Backward propagation recovers the initial state without frame, epoch, or unit drift.		

Acceptance Criteria	The STK results are taken as the true value. For the nominal case, the reference-state difference shall not exceed 0.010 km in position or 1.0e-5 km/s in velocity over 24 h. Relative drift of energy and angular-momentum magnitude shall not exceed 1.0e-9. All edge cases shall remain finite and deterministic.		
Case Designer		Executor / Date	
Actual Results	Not Executed. Attach configuration manifest, input hash, output file, comparison report, and relevant log excerpts during formal execution.		
Recorded By		Test Conclusion	☐ Pass ☐ Fail ☒ Not Executed
Anomalies	None recorded; case not executed.		

3.1.2 Test Project 2: J2

This project verifies the J2-Perturbed Orbit Propagation capability of the Space Data Center digital twin orbit-dynamics module.

Table 3. OD-002 - J2-Perturbed Orbit Propagation Test Case

Case ID	OD-002	Case Name	J2-Perturbed Orbit Propagation
Test Objective	Verify the oblateness perturbation, including acceleration direction and the secular change of the orbital plane and apsides.		
Preconditions	- The same central-body constants are used by the system under test and the independent J2 reference integrator. - Only point-mass gravity and J2 are enabled; drag, third-body gravity, and radiation pressure are disabled.		
Input Data	- WGS-84-compatible Earth radius and the configured J2 coefficient, recorded in the run manifest. - Sun-synchronous-like LEO and low-inclination LEO cases propagated for 24 h with 30 s or finer integration output.		
Procedure	- Run a two-body baseline and then enable J2 without changing the initial state or integrator settings. - Compare J2 acceleration vectors at representative equatorial, polar, and inclined positions with the analytic expression. - Propagate each orbit and compare sampled states with an independently configured J2 Cowell integrator. - Estimate RAAN and argument-of-periapsis rates and compare their direction and magnitude with first-order secular theory.		

Expected Results	• The J2 acceleration has the correct sign, symmetry, and distance scaling. • The propagated orbit exhibits the expected nodal and apsidal precession. • Disabling J2 returns the two-body baseline within numerical tolerance.		
Acceptance Criteria	The STK results are taken as the true value. Acceleration components shall agree with the analytic J2 equation to a relative tolerance of 1.0e-10. State differences against the identically configured reference shall not exceed 0.100 km in position or 1.0e-4 km/s in velocity at 24 h. Secular-rate differences shall not exceed 0.05 deg/day.		
Case Designer		Executor / Date	
Actual Results	Not Executed. Attach configuration manifest, input hash, output file, comparison report, and relevant log excerpts during formal execution.		
Recorded By		Test Conclusion	☐ Pass ☐ Fail ☒ Not Executed
Anomalies	None recorded; case not executed.		

3.1.3 Test Project 3: SGP4

This project verifies the SGP4 TLE Propagation capability of the Space Data Center digital twin orbit-dynamics module.

Table 4. OD-003 - SGP4 TLE Propagation Test Case

Case ID	OD-003	Case Name	SGP4 TLE Propagation
Test Objective	Verify TLE parsing and SGP4 propagation against validated SGP4 reference vectors without treating HPOP as the oracle.		
Preconditions	• The implementation uses the configured SGP4 gravity constants, normally WGS-72 for standard TLE compatibility. • Validated near-Earth and deep-space TLE test vectors are available with their expected TEME states.		
Input Data	• Representative LEO, MEO, GEO, high-eccentricity, and decayed/near-limit TLEs, including checksums and epochs. • Evaluation times spanning at least -1440 to +1440 minutes from each TLE epoch.		

Procedure	1. Parse each TLE and verify line number, catalog number, epoch, checksum handling, and unit conversion. 2. Propagate to every reference time and export the TEME position, TEME velocity, and status code. 3. Compare the output with a validated independent SGP4 implementation using the same constants. 4. Inject malformed TLEs and out-of-range propagation times to verify explicit error reporting.		
Expected Results	• Valid TLEs reproduce the validated SGP4 state vectors and deep-space branch behavior. • Invalid input is rejected without returning a plausible but unqualified state. • Output metadata clearly identifies TEME, the epoch, time scale, and propagation status.		
Acceptance Criteria	The STK results are taken as the true value. For validated vectors, the maximum difference shall not exceed 0.001 km in position or 1.0e-6 km/s in velocity. Expected SGP4 error codes shall match the reference. All malformed-input tests shall produce deterministic validation errors.		
Case Designer		Executor / Date	
Actual Results	Not Executed. Attach configuration manifest, input hash, output file, comparison report, and relevant log excerpts during formal execution.		
Recorded By		Test Conclusion	☐ Pass ☐ Fail ☒ Not Executed
Anomalies	None recorded; case not executed.		

3.1.4 Test Project 4: HPOP

This project verifies the High-Precision Orbit Propagation capability of the Space Data Center digital twin orbit-dynamics module.

Table 5. OD-004 - High-Precision Orbit Propagation Test Case

Case ID	OD-004	Case Name	High-Precision Orbit Propagation
Test Objective	Verify high-precision numerical propagation, force-model configuration, force contribution toggles, and reproducibility.		
Preconditions	• Gravity field, atmosphere, space-weather, ephemeris, Earth-orientation, drag, SRP, and third-body files are versioned and hashed. • An independent HPOP-capable reference is configured with the same constants, frames, force models, and integrator tolerances.		

Input Data	• LEO, MEO, GEO, and high-eccentricity initial states with spacecraft mass, drag coefficient, reflectivity coefficient, and areas. • Force-model configurations ranging from central gravity only to the project operational configuration; spans of 24 h and 7 days.		
Procedure	1. Run the central-gravity baseline and record integrator step statistics and state outputs. 2. Enable geopotential, third bodies, drag, tides if supported, and SRP one at a time; confirm that each contribution is visible in acceleration diagnostics. 3. Run the full configuration twice with identical inputs and compare outputs bitwise or within the documented deterministic tolerance. 4. Compare sampled states and event times with the identically configured independent reference.		
Expected Results	• Every enabled force has the correct direction, order of magnitude, and configuration provenance. • The full trajectory is continuous, reproducible, and accompanied by an auditable force-model manifest. • No silent fallback occurs when a required data file is missing or outside its validity interval.		
Acceptance Criteria	The STK results are taken as the true value. For the agreed 24 h LEO benchmark, the independent-reference difference shall not exceed 1.0 km in position or 0.001 km/s in velocity unless the approved model-specific tolerance is stricter. Repeat runs shall be deterministic. Missing or expired auxiliary data shall generate a blocking diagnostic.		
Case Designer		Executor / Date	
Actual Results	Not Executed. Attach configuration manifest, input hash, output file, comparison report, and relevant log excerpts during formal execution.		
Recorded By		Test Conclusion	☐ Pass ☐ Fail ☒ Not Executed
Anomalies	None recorded; case not executed.		

3.1.5 Test Project 5: OE2RV

This project verifies the Orbital Elements to Position and Velocity capability of the Space Data Center digital twin orbit-dynamics module.

Table 6. CV-001 - Orbital Elements to Position and Velocity Test Case

Case ID	CV-001	Case Name	Orbital Elements to Position and Velocity
Test Objective	Verify conversion from orbital elements to Cartesian state with correct units, quadrants, rotation order, and singular-case behavior.		

Preconditions	- The supported element set, anomaly type, central-body parameter, inertial frame, and angle units are explicitly configured. - An independent conversion routine is available for vector-level comparison.		
Input Data	- Nominal elliptic, circular, equatorial, polar, retrograde, and high-eccentricity element sets. - Boundary values near 0/360 deg and near the implementation's eccentricity and inclination limits.		
Procedure	- Convert every element set to position and velocity using the specified anomaly convention. - Compare the vectors with an independent rotation-matrix implementation. - Convert each state back through RV2OE and reconstruct a second Cartesian state. - Verify explicit handling of invalid semi-major axis, eccentricity, non-finite values, and undefined classical angles.		
Expected Results	- Nominal conversions match the independent vectors and preserve orbital energy and angular momentum. - Circular/equatorial ambiguity is handled by the documented convention rather than random angle selection. - Invalid inputs are rejected with field-specific messages.		
Acceptance Criteria	The STK results are taken as the true value. For valid nonsingular cases, vector differences shall not exceed 1.0e-8 km in position or 1.0e-10 km/s in velocity. For singular cases, the reconstructed Cartesian state shall meet the same tolerance even when individual classical angles are not unique.		
Case Designer		Executor / Date	
Actual Results	Not Executed. Attach configuration manifest, input hash, output file, comparison report, and relevant log excerpts during formal execution.		
Recorded By		Test Conclusion	[] Pass [] Fail [X] Not Executed
Anomalies	None recorded; case not executed.		

3.1.6 Test Project 6: RV2OE

This project verifies the Position and Velocity to Orbital Elements capability of the Space Data Center digital twin orbit-dynamics module.

Table 7. CV-002 - Position and Velocity to Orbital Elements Test Case

Case ID	CV-002	Case Name	Position and Velocity to Orbital Elements
----------------	---------------	------------------	--

Test Objective	Verify conversion from Cartesian state to orbital elements, including quadrant resolution and well-defined singular-case outputs.		
Preconditions	• Frame, epoch, gravitational parameter, element ordering, anomaly type, and angle normalization range are fixed. • OE2RV and an independent vector-based oracle are available for round-trip testing.		
Input Data	• Cartesian states representing nominal elliptic, circular, equatorial, polar, retrograde, and high-eccentricity orbits. • Invalid states including zero position, non-finite components, and physically unsupported energy/eccentricity ranges.		
Procedure	1. Convert each valid Cartesian state to the supported orbital-element representation. 2. Check semi-major axis, eccentricity, inclination, node, periapsis, and anomaly against the independent oracle. 3. Convert the reported elements through OE2RV and compare the reconstructed state with the input. 4. Verify angle normalization and the documented alternative angles for circular/equatorial cases.		
Expected Results	• Quadrants and angle wrapping are correct for all orientations. • Round-trip reconstruction preserves the input state even when classical elements are non-unique. • Undefined or invalid cases return an explicit status and do not emit NaN values without explanation.		
Acceptance Criteria	The STK results are taken as the true value. For valid cases, reconstructed-state differences shall not exceed 1.0e-8 km in position or 1.0e-10 km/s in velocity. Nonsingular element differences shall not exceed 1.0e-10 in dimensionless values and 1.0e-8 rad in angles after normalization.		
Case Designer		Executor / Date	
Actual Results	Not Executed. Attach configuration manifest, input hash, output file, comparison report, and relevant log excerpts during formal execution.		
Recorded By		Test Conclusion	[] Pass [] Fail [X] Not Executed
Anomalies	None recorded; case not executed.		

3.1.7 Test Project 7: ECI2ECEF

This project verifies the ECI to ECEF Coordinate Transformation capability of the Space Data

Center digital twin orbit-dynamics module.

Table 8. CV-003 - ECI to ECEF Coordinate Transformation Test Case

Case ID	CV-003	Case Name	ECI to ECEF Coordinate Transformation
Test Objective	Verify epoch-dependent transformation of position and velocity from the configured inertial frame to the configured Earth-fixed frame.		
Preconditions	- The exact ECI and ECEF realizations, time scale, Earth-orientation model, leap-second table, and EOP data source are recorded. - An independent standards-compatible transformation is available for reference values.		
Input Data	- Reference epochs covering different sidereal angles, leap-second boundaries where supported, and normal/EOP-missing conditions. - Axis-aligned vectors, representative orbital states, and known ground-station ECEF coordinates.		
Procedure	- Transform position-only vectors and verify the rotation matrix orthogonality and determinant. - Transform full position/velocity states and confirm inclusion of the Earth-rotation transport term. - Compare results with the independent reference for all epochs. - Apply the inverse transformation and compare the recovered inertial state with the input.		
Expected Results	- The transformation uses the requested frames and epoch, not a hard-coded Greenwich angle. - Position and velocity are consistent with Earth rotation and the configured EOP fidelity. - Missing or out-of-range Earth-orientation data is surfaced according to policy.		
Acceptance Criteria	The STK results are taken as the true value. With identical EOP and frame definitions, reference differences shall not exceed 0.001 km in position or 1.0e-6 km/s in velocity. The rotation matrix shall satisfy $R \cdot R^T = I$ within 1.0e-12, and inverse round-trip state errors shall meet the same state tolerances.		
Case Designer		Executor / Date	
Actual Results	Not Executed. Attach configuration manifest, input hash, output file, comparison report, and relevant log excerpts during formal execution.		
Recorded By		Test Conclusion	☐ Pass ☐ Fail ☒ Not Executed
Anomalies	None recorded; case not executed.		

3.2 Attitude Kinematics and Dynamics

3.2.1 Test Project 8: YPR2EULER

This project verifies the Yaw-Pitch-Roll to Euler-Angle Conversion capability of the Space Data Center digital twin orbit-dynamics module.

Table 9. AD-001 - Yaw-Pitch-Roll to Euler-Angle Conversion Test Case

Case ID	AD-001	Case Name	Yaw-Pitch-Roll to Euler-Angle Conversion
Test Objective	Verify conversion between the documented yaw-pitch-roll convention and the requested Euler sequence using rotation-equivalent comparisons.		
Preconditions	• Active/passive rotation sense, body/reference frames, axis sequence, multiplication order, handedness, and angle units are explicitly documented. • A rotation-matrix or quaternion oracle is available.		
Input Data	• Zero attitude; positive and negative single-axis rotations; combined rotations; wraparound cases; and samples close to gimbal lock. • All Euler sequences supported by the module, with the primary 3-2-1 sequence tested at minimum.		
Procedure	1. Convert each yaw-pitch-roll input to Euler angles using the selected sequence. 2. Construct rotation matrices independently from the input and output and compare them element by element. 3. Verify output angle ranges, continuity away from singularities, and deterministic wrap behavior. 4. Approach and cross the singularity to verify the documented gimbal-lock status and fallback convention.		
Expected Results	• Input and output represent the same physical attitude for every nonsingular case. • No hidden degree/radian conversion or axis-order reversal occurs. • Singular cases are flagged and returned using a deterministic documented convention.		
Acceptance Criteria	The STK results are taken as the true value. The maximum absolute rotation-matrix element difference shall not exceed $1.0\text{e-}10$, and the quaternion angular separation shall not exceed $1.0\text{e-}8$ rad for nonsingular cases. Singular cases shall return the expected status without NaN or discontinuous random branches.		
Case Designer		Executor / Date	
Actual Results	Not Executed. Attach configuration manifest, input hash, output file, comparison report, and relevant log excerpts during formal execution.		

Recorded By		Test Conclusion	[] Pass [] Fail [X] Not Executed
Anomalies	None recorded; case not executed.		

3.2.2 Test Project 9: EULER2YPR

This project verifies the Yaw-Pitch-Roll to Euler-Angle Conversion capability of the Space Data Center digital twin orbit-dynamics module.

Table 9. AD-001 - Yaw-Pitch-Roll to Euler-Angle Conversion Test Case

Case ID	AD-001	Case Name	Yaw-Pitch-Roll to Euler-Angle Conversion
Test Objective	Verify conversion between the documented yaw-pitch-roll convention and the requested Euler sequence using rotation-equivalent comparisons.		
Preconditions	- Active/passive rotation sense, body/reference frames, axis sequence, multiplication order, handedness, and angle units are explicitly documented. - A rotation-matrix or quaternion oracle is available.		
Input Data	- Zero attitude; positive and negative single-axis rotations; combined rotations; wraparound cases; and samples close to gimbal lock. - All Euler sequences supported by the module, with the primary 3-2-1 sequence tested at minimum.		
Procedure	- Convert each yaw-pitch-roll input to Euler angles using the selected sequence. - Construct rotation matrices independently from the input and output and compare them element by element. - Verify output angle ranges, continuity away from singularities, and deterministic wrap behavior. - Approach and cross the singularity to verify the documented gimbal-lock status and fallback convention.		
Expected Results	- Input and output represent the same physical attitude for every nonsingular case. - No hidden degree/radian conversion or axis-order reversal occurs. - Singular cases are flagged and returned using a deterministic documented convention.		
Acceptance Criteria	The STK results are taken as the true value. The maximum absolute rotation-matrix element difference shall not exceed 1.0e-10, and the quaternion angular separation shall not exceed 1.0e-8 rad for nonsingular cases. Singular cases shall return the expected status without NaN or discontinuous random branches.		
Case Designer		Executor / Date	

Actual Results	Not Executed. Attach configuration manifest, input hash, output file, comparison report, and relevant log excerpts during formal execution.		
Recorded By		Test Conclusion	[] Pass [] Fail [X] Not Executed
Anomalies	None recorded; case not executed.		

3.2.3 Test Project 10: EULERDYNAMICS

This project verifies the Rigid-Body Euler Dynamics capability of the Space Data Center digital twin orbit-dynamics module.

Table 10. AD-002 - Rigid-Body Euler Dynamics Test Case

Case ID	AD-002	Case Name	Rigid-Body Euler Dynamics
Test Objective	Verify Euler rigid-body rotational dynamics, torque application, attitude integration, and conservation properties.		
Preconditions	- The inertia tensor is symmetric, positive definite, and expressed in the documented body frame. - Torque units, body/reference expression frame, attitude representation, and integrator settings are recorded.		
Input Data	- Spherical, axisymmetric, and asymmetric inertia tensors with principal-axis and general initial angular velocities. - Torque-free cases, constant principal-axis torque, and a time-varying torque profile.		
Procedure	- Run torque-free principal-axis rotation and compare angular rate with the analytic constant-rate solution. - Run a torque-free asymmetric case and monitor inertial angular momentum and rotational kinetic energy. - Apply constant torque about a principal axis and compare angular acceleration with torque divided by principal inertia. - Verify quaternion or direction-cosine-matrix normalization throughout long-duration integration.		
Expected Results	- Euler's equations are satisfied with correct cross-product signs and torque frame handling. - Torque-free invariants are conserved within integration tolerance. - Attitude remains normalized and continuous without nonphysical energy growth.		

Acceptance Criteria	The STK results are taken as the true value. Principal-axis angular-rate error shall not exceed 1.0e-8 rad/s. Relative drift of torque-free kinetic energy and inertial angular-momentum magnitude shall not exceed 1.0e-6 over the test span. Attitude normalization error shall not exceed 1.0e-10.		
Case Designer		Executor / Date	
Actual Results	Not Executed. Attach configuration manifest, input hash, output file, comparison report, and relevant log excerpts during formal execution.		
Recorded By		Test Conclusion	☐ Pass ☐ Fail ☒ Not Executed
Anomalies	None recorded; case not executed.		

3.3 Sun, Radiation, Thermal, and Power

3.3.1 Test Project 11: SUN POSITION ANALYTICAL

This project verifies the Analytical Sun Position capability of the Space Data Center digital twin orbit-dynamics module.

Table 11. SE-001 - Analytical Sun Position Test Case

Case ID	SE-001	Case Name	Analytical Sun Position
Test Objective	Verify the analytical Sun-position model, frame/epoch metadata, annual continuity, and its stated accuracy envelope.		
Preconditions	- The analytical formulation, valid date range, time scale, and output frame are documented. - The ephemeris-based Sun position module is available as the comparison oracle.		
Input Data	- Daily epochs over one full year plus solstices, equinoxes, and the valid-range boundaries. - Requested outputs as unit direction, distance, and Cartesian vector where supported.		
Procedure	- Evaluate the analytical Sun position at all epochs and confirm continuity across day/year boundaries. - Compute the corresponding ephemeris-based Sun vector in the same frame and time scale. - Compare angular separation, distance, and Cartesian direction trends. - Request epochs outside the documented validity interval and verify the warning/error policy.		
Expected Results	- The analytical vector follows the correct annual direction and distance trend. - Frame, origin, epoch, and time scale are attached to every result. - Out-of-validity use is not silently reported at nominal accuracy.		

Acceptance Criteria	The STK results are taken as the true value. Within the documented valid interval, angular separation from the ephemeris direction shall not exceed 0.5 deg and relative Sun-distance error shall not exceed 0.5%, unless the approved model specification defines stricter limits. No discontinuity greater than the expected physical step shall occur.		
Case Designer		Executor / Date	
Actual Results	Not Executed. Attach configuration manifest, input hash, output file, comparison report, and relevant log excerpts during formal execution.		
Recorded By		Test Conclusion	☐ Pass ☐ Fail ☒ Not Executed
Anomalies	None recorded; case not executed.		

3.3.2 Test Project 12: SUN POSITION EPHEMERIS

This project verifies the Ephemeris-Based Sun Position capability of the Space Data Center digital twin orbit-dynamics module.

Table 12. SE-002 - Ephemeris-Based Sun Position Test Case

Case ID	SE-002	Case Name	Ephemeris-Based Sun Position
Test Objective	Verify ephemeris loading, interpolation, time/frame conversion, boundary handling, and reproducible Sun vectors.		
Preconditions	- The approved planetary ephemeris file and its hash, coverage interval, interpolation method, and time scale are recorded. - Trusted reference vectors from the same ephemeris are available at knots and intermediate epochs.		
Input Data	- Reference epochs at ephemeris records, between records, and just inside/outside the coverage interval. - Multiple requested output frames supported by the module.		
Procedure	- Load the ephemeris and verify file identity, coverage, and metadata diagnostics. - Evaluate Sun position and velocity at reference and interpolation epochs. - Compare results with the trusted vectors after applying identical frame and origin definitions. - Test missing, corrupted, and out-of-range ephemeris inputs.		

Expected Results	- Position and velocity are continuous across record boundaries and match the selected ephemeris. - Frame and origin transformations do not alter vector meaning. - Unavailable data produces an explicit blocking error rather than an undocumented fallback.		
Acceptance Criteria	The STK results are taken as the true value. Using the same ephemeris and frame definitions, position difference shall not exceed 10 km and velocity difference shall not exceed 1.0e-5 km/s, or the ephemeris provider's tighter interpolation tolerance. All boundary and corrupted-file tests shall return the documented status.		
Case Designer		Executor / Date	
Actual Results	Not Executed. Attach configuration manifest, input hash, output file, comparison report, and relevant log excerpts during formal execution.		
Recorded By		Test Conclusion	☐ Pass ☐ Fail ☒ Not Executed
Anomalies	None recorded; case not executed.		

3.3.3 Test Project 13: SUN BETA ANGLE

This project verifies the Sun Beta-Angle Calculation capability of the Space Data Center digital twin orbit-dynamics module.

Table 13. SE-003 - Sun Beta-Angle Calculation Test Case

Case ID	SE-003	Case Name	Sun Beta-Angle Calculation
Test Objective	Verify the signed angle between the Sun direction and the orbital plane, including sign convention and near-zero cases.		
Preconditions	- The orbit-normal direction, beta sign convention, input frames, and angle range are documented. - Sun and spacecraft state vectors can be evaluated at a common epoch and frame.		
Input Data	- Constructed vectors for beta angles of 0, +/-30, +/-60, and near +/-90 deg. - Representative LEO states with analytical and ephemeris Sun vectors.		
Procedure	1. Compute the orbit normal from position cross velocity and normalize all vectors. 2. Calculate beta independently from the signed dot product and compare with the module output. 3. Reverse orbit direction and verify the documented sign change. 4. Exercise nearly coplanar and nearly normal Sun directions to check numerical clamping.		

Expected Results	• The reported beta angle has the documented sign and range. • The result is invariant to input vector magnitudes and consistent across equivalent frames. • Floating-point roundoff near +/-1 dot products does not produce invalid inverse-trigonometric input.		
Acceptance Criteria	The STK results are taken as the true value. The absolute beta-angle difference from the independent vector calculation shall not exceed 0.01 deg for all valid cases. Constructed boundary cases shall remain finite, correctly signed, and within the documented angle range.		
Case Designer		Executor / Date	
Actual Results	Not Executed. Attach configuration manifest, input hash, output file, comparison report, and relevant log excerpts during formal execution.		
Recorded By		Test Conclusion	☐ Pass ☐ Fail ☒ Not Executed
Anomalies	None recorded; case not executed.		

3.3.4 Test Project 14: SUN INTENSITY

This project verifies the Solar Intensity and Eclipse Factor capability of the Space Data Center digital twin orbit-dynamics module.

Table 14. SE-004 - Solar Intensity and Eclipse Factor Test Case

Case ID	SE-004	Case Name	Solar Intensity and Eclipse Factor
Test Objective	Verify inverse-square solar intensity, incidence-independent irradiance output, and configured eclipse/penumbra attenuation.		
Preconditions	• The reference solar constant, astronomical unit, eclipse model, central-body radius, and units are recorded. • Sun-spacecraft distance and shadow geometry can be supplied independently.		
Input Data	• Sun distances of 0.983, 1.000, and 1.017 AU; full-sun, umbra, and penumbra geometries. • Boundary samples immediately before, during, and after eclipse transitions.		
Procedure	1. Evaluate full-sun intensity at each distance and compare with $S = S_0 \cdot (1 \text{ AU}/r)^2$. 2. Evaluate constructed umbra and penumbra cases and record the illumination factor. 3. Sweep through eclipse entry and exit to verify bounded, monotonic transition behavior. 4. Test invalid distance and non-finite vector inputs.		

Expected Results	• Full-sun intensity follows the inverse-square law using the configured constant. • The illumination factor remains within [0, 1], with the correct full-sun and full-umbra limits. • Invalid geometry is rejected rather than producing negative or unbounded irradiance.		
Acceptance Criteria	The STK results are taken as the true value. Full-sun intensity shall agree with the independent equation within 0.1%. Illumination factor shall be 1 in full sun and 0 in full umbra for a hard-shadow model; penumbra values shall remain bounded and monotonic. Output units shall be W/m ² .		
Case Designer		Executor / Date	
Actual Results	Not Executed. Attach configuration manifest, input hash, output file, comparison report, and relevant log excerpts during formal execution.		
Recorded By		Test Conclusion	☐ Pass ☐ Fail ☒ Not Executed
Anomalies	None recorded; case not executed.		

3.3.5 Test Project 15: POWER CONSIDERING THERMAL

This project verifies the Solar Power with Thermal Effects capability of the Space Data Center digital twin orbit-dynamics module.

Table 15. SE-005 - Solar Power with Thermal Effects Test Case

Case ID	SE-005	Case Name	Solar Power with Thermal Effects
Test Objective	Verify solar-array power calculation with incidence, illumination, temperature coefficient, area, and efficiency limits.		
Preconditions	• Panel area, normal vector, reference efficiency, reference temperature, temperature coefficient, wiring/conversion efficiency, and clipping policy are configured. • Solar intensity and panel temperature can be injected as controlled inputs.		
Input Data	• Incidence angles of 0, 30, 60, 90, and 120 deg; illumination factors of 0, 0.5, and 1. • Panel temperatures below, at, and above the reference temperature, including configured operating limits.		
Procedure	1. Calculate the independent expected power from irradiance, projected area, reference efficiency, thermal coefficient, and downstream efficiency. 2. Sweep incidence angle while holding all other inputs constant. 3. Sweep panel temperature and verify the configured linear or nonlinear thermal response. 4. Test eclipse, back-side incidence, invalid area, and out-of-range temperature handling.		

Expected Results	• Power decreases with cosine incidence and follows the documented temperature coefficient. • Eclipse and back-side illumination do not generate negative or nonphysical power. • Limits, clipping, and thermal-model provenance are reported with the output.		
Acceptance Criteria	The STK results are taken as the true value. For all valid cases, electrical power shall agree with the independent equation within 0.5%. Power shall be zero for zero illumination and for incidence at or beyond 90 deg unless a documented back-side model is enabled. No result shall exceed the documented physical/configuration bounds.		
Case Designer		Executor / Date	
Actual Results	Not Executed. Attach configuration manifest, input hash, output file, comparison report, and relevant log excerpts during formal execution.		
Recorded By		Test Conclusion	☐ Pass ☐ Fail ☒ Not Executed
Anomalies	None recorded; case not executed.		

3.3.6 Test Project 16: SUN RADIATION PRESSURE

This project verifies the Solar Radiation Pressure Acceleration capability of the Space Data Center digital twin orbit-dynamics module.

Table 16. SE-006 - Solar Radiation Pressure Acceleration Test Case

Case ID	SE-006	Case Name	Solar Radiation Pressure Acceleration
Test Objective	Verify solar-radiation-pressure magnitude, direction, distance scaling, area-to-mass scaling, reflectivity, and eclipse gating.		
Preconditions	• The SRP model type, reference pressure/intensity, speed of light, reflectivity coefficient, illuminated area model, mass, and shadow model are recorded. • A cannonball-model oracle is available for the baseline test.		
Input Data	• Multiple Sun-spacecraft directions and distances; spacecraft masses, areas, and reflectivity coefficients. • Full-sun, penumbra, and umbra illumination factors plus invalid mass/area inputs.		
Procedure	1. Evaluate baseline cannonball SRP and compare magnitude with $P_0 \cdot (1 \text{ AU}/r)^2 \cdot C_r \cdot A/m$ times illumination. 2. Rotate the Sun direction and verify the acceleration direction in the requested frame. 3. Vary mass, area, reflectivity, and distance independently to verify proportional scaling. 4. Enter eclipse and confirm that the acceleration follows the same illumination factor as the configured shadow model.		

Expected Results	• SRP points away from the Sun for the documented force convention and scales with $C_r \cdot A/m$ and inverse-square distance. • Umbra suppresses SRP and penumbra applies bounded partial attenuation. • Invalid physical parameters are rejected.		
Acceptance Criteria	The STK results are taken as the true value. Baseline acceleration magnitude shall agree with the independent cannonball equation within 0.1%, and direction angular error shall not exceed $1.0e-6$ rad. Full-umbra acceleration shall be zero within $1.0e-15$ km/s ² . Scaling tests shall match the expected ratios within 0.1%.		
Case Designer		Executor / Date	
Actual Results	Not Executed. Attach configuration manifest, input hash, output file, comparison report, and relevant log excerpts during formal execution.		
Recorded By		Test Conclusion	☐ Pass ☐ Fail ☒ Not Executed
Anomalies	None recorded; case not executed.		

3.4 Access Analysis

3.4.1 Test Project 17: ACCESS PROPAGATION

This project verifies the Access Interval Propagation capability of the Space Data Center digital twin orbit-dynamics module.

Table 17. AC-001 - Access Interval Propagation Test Case

Case ID	AC-001	Case Name	Access Interval Propagation
Test Objective	Verify propagation-based access detection, line-of-sight geometry, constraint application, event bracketing, and interval reporting.		
Preconditions	• Satellite propagator, target/ground-station coordinates, Earth model, frame conversions, and access constraints are fixed. • An independent dense-sampling or validated access-analysis reference is available.		
Input Data	• LEO and MEO satellites; ground stations at low, mid, and high latitude; minimum elevation masks of 0, 5, and 10 deg. • Intervals containing no pass, one pass, multiple passes, a pass crossing the scenario boundary, and a grazing pass.		

Procedure	1. Propagate satellite and target states and compute line-of-sight/elevation at dense reference samples. 2. Run ACCESS PROPAGATION over the same interval and collect access start, stop, duration, maximum elevation, and status. 3. Compare all reported intervals with the independent reference and inspect event-function signs around each boundary. 4. Repeat with masks and scenario boundaries changed to verify clipping, merging, and no-access behavior.		
Expected Results	• All and only geometrically valid access intervals are reported in chronological non-overlapping order. • Constraint and scenario-boundary behavior is explicit and deterministic. • Event metadata identifies the target, frame, mask, start/stop epochs, and time scale.		
Acceptance Criteria	The STK results are taken as the true value. No valid reference interval shall be missed and no false interval shall be added. Start and stop epoch differences shall not exceed 1.0 s, and duration difference shall not exceed 2.0 s. Elevation at an unconstrained event boundary shall equal the configured mask within 0.01 deg.		
Case Designer		Executor / Date	
Actual Results	Not Executed. Attach configuration manifest, input hash, output file, comparison report, and relevant log excerpts during formal execution.		
Recorded By		Test Conclusion	[] Pass [] Fail [X] Not Executed
Anomalies	None recorded; case not executed.		

3.4.2 Test Project 18: ACCESS INTERPOLATION

This project verifies the Access State and Event Interpolation capability of the Space Data Center digital twin orbit-dynamics module.

Table 18. AC-002 - Access State and Event Interpolation Test Case

Case ID	AC-002	Case Name	Access State and Event Interpolation
Test Objective	Verify interpolation of access geometry and event epochs from sampled propagation data without missed or spurious intervals.		
Preconditions	• Interpolation order, sample cadence limits, bracketing rules, extrapolation policy, and event tolerance are configured. • Dense propagation output is available as the reference truth set.		

Input Data	• Reference access scenarios from AC-001 sampled at 1, 5, 10, and 30 s cadences. • Smooth passes, short passes, grazing events, duplicate timestamps, gaps, out-of-order samples, and scenario-edge events.		
Procedure	1. Generate dense reference geometry and event epochs, then down-sample to each supported cadence. 2. Run ACCESS INTERPOLATION to reconstruct geometry and start/stop events from the sampled data. 3. Compare interpolated range/elevation and event epochs with dense reference values. 4. Inject gaps, duplicates, out-of-order data, and extrapolation requests to verify validation and quality flags.		
Expected Results	• Within the documented cadence envelope, interpolation is continuous and preserves the access/no-access topology. • Event roots remain bracketed and accurate; insufficient data is reported with a quality/status flag. • The module does not extrapolate or bridge gaps unless explicitly configured to do so.		
Acceptance Criteria	The STK results are taken as the true value. For supported sample cadences, no reference interval shall be missed or added. Interpolated start/stop epochs shall be within 0.2 s of dense propagation and elevation within 0.01 deg. Invalid sample sequences and unsupported gaps/extrapolation shall return deterministic validation errors or quality flags.		
Case Designer		Executor / Date	
Actual Results	Not Executed. Attach configuration manifest, input hash, output file, comparison report, and relevant log excerpts during formal execution.		
Recorded By		Test Conclusion	[] Pass [] Fail [X] Not Executed
Anomalies	None recorded; case not executed.		

4 Test Content Sufficiency Analysis

The test set provides one-to-one functional coverage for all 17 requested capabilities. It also covers the principal cross-cutting risks: frame and time-system ambiguity, units, model-data provenance, numerical tolerance, deterministic failure behavior, singular geometry, edge conditions, and independent-reference comparison.

Coverage is specification-complete but not execution-complete. Formal closure requires a controlled build, approved configuration/data files, an independent oracle, captured outputs, and reviewer acceptance for every case.

5 Test Conditions and Requirements

- Record product version, source commit, build type, compiler/runtime, operating system, processor architecture, and enabled acceleration libraries.
- Record the unit system, epoch/time scale, inertial and Earth-fixed frame realizations, and all angle conventions.
- Freeze and hash gravity, atmosphere, space-weather, EOP, leap-second, planetary ephemeris, thermal, spacecraft, and access-constraint data.
- Use deterministic seeds and thread settings where parallel execution could change floating-point reduction order.
- Synchronize clocks and retain service/interface logs for distributed digital-twin tests.
- Run reference calculations independently; do not use one module under test as the sole oracle for another module with shared code.

6 Test Data

All test data shall be versioned and accompanied by origin, validity interval, frame/time metadata, and a cryptographic hash.

Table 19. Test Data Inventory

No.	Test Data	Data Type	Form	Source
1	Initial orbit states and element sets	Input	JSON/CSV	Project test dataset
2	Validated TLE and SGP4 vectors	Input / Oracle	TLE/TXT	Approved SGP4 validation set
3	Gravity, atmosphere, space-weather, and EOP data	Model data	Native files	Approved data repository

No.	Test Data	Data Type	Form	Source
4	Planetary ephemeris and Sun reference vectors	Model / Oracle	Binary/CSV	Approved ephemeris source
5	Spacecraft mass, area, optical, thermal, and power parameters	Input	JSON/CSV	Spacecraft configuration baseline
6	Ground stations, targets, masks, and access truth intervals	Input / Oracle	JSON/CSV	Project access test dataset
7	Run manifests, module outputs, logs, and comparison reports	Evidence	JSON/CSV/XT T	Generated during execution

7 Test Summary

All cases are intentionally marked Not Executed. Replace the status only after the acceptance criteria have been evaluated against retained execution evidence.

Table 20. Orbit Dynamics Module Test Summary

No.	Case ID	Function	Verification Focus	Status	Result / Evidence
1	OD-001	TWOBODY	Keplerian accuracy, invariants, reversibility	Not Executed	To be completed with run manifest, output, comparison report, and logs.
2	OD-002	J2	J2 acceleration and secular precession	Not Executed	To be completed with run manifest, output, comparison report, and logs.
3	OD-003	SGP4	TLE parsing, TEME states, validated SGP4 vectors	Not Executed	To be completed with run manifest, output, comparison report, and logs.
4	OD-004	HPOP	Force-model fidelity, reproducibility, auxiliary data	Not Executed	To be completed with run manifest, output, comparison report, and logs.
5	CV-001	OE2RV	Element conventions, rotations, singularities	Not Executed	To be completed with run manifest, output, comparison report, and logs.
6	CV-002	RV2OE	Quadrants, angle normalization, round-trip state	Not Executed	To be completed with run manifest, output, comparison report, and logs.
7	CV-003	ECI2ECEF	Frame/time fidelity and velocity transport term	Not Executed	To be completed with run manifest, output, comparison report, and logs.
8	AD-001	YPR2EULER	Rotation convention, sequence, gimbal lock	Not Executed	To be completed with run manifest, output, comparison report, and logs.
9	AD-002	EULERDYNAMICS	Euler equations, torque response, conserved quantities	Not Executed	To be completed with run manifest, output, comparison report, and logs.
10	SE-001	SUN POSITION ANALYTICAL	Analytical accuracy envelope and continuity	Not Executed	To be completed with run manifest, output, comparison report, and logs.
11	SE-002	SUN POSITION EPHEMERIS	Ephemeris identity, interpolation, coverage	Not Executed	To be completed with run manifest, output, comparison report, and logs.
12	SE-003	SUN BETA ANGLE	Signed orbit-plane geometry and numerical	Not Executed	To be completed with run manifest, output, comparison report, and logs.

No.	Case ID	Function	Verification Focus	Status	Result / Evidence
			bounds		
13	SE-004	SUN INTENSITY	Inverse-square irradiance and eclipse attenuation	Not Executed	To be completed with run manifest, output, comparison report, and logs.
14	SE-005	POWER CONSIDERING THERMAL	Incidence, temperature coefficient, physical clipping	Not Executed	To be completed with run manifest, output, comparison report, and logs.
15	SE-006	SUN RADIATION PRESSURE	SRP magnitude, direction, scaling, eclipse gating	Not Executed	To be completed with run manifest, output, comparison report, and logs.
16	AC-001	ACCESS PROPAGATION	LOS constraints, event detection, interval boundaries	Not Executed	To be completed with run manifest, output, comparison report, and logs.
17	AC-002	ACCESS INTERPOLATION	Interpolation error, event topology, data-quality handling	Not Executed	To be completed with run manifest, output, comparison report, and logs.

Overall Conclusion: ☐ Pass ☐ Fail ☒ Not Executed

On-site Test Expert Signature: _____ Date: _____